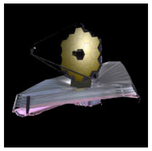

10675 - The Nearest Seyfert Starburst Nucleus and its Galactic Outflow

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
NGC4945 MIRI Imaging				
	1	MIRI Imaging Outflows	MIRI Imaging	(1) NGC-4945
	2	MIRI Imaging Center	MIRI Imaging	(1) NGC-4945
NGC4945 MIRI MRS				
	3	MIRI MRS Center	MIRI Medium Resolution Spectroscopy	(1) NGC-4945
	4	MRS Background	MIRI Medium Resolution Spectroscopy	(2) NGC4945-off
NGC4945 NIRCam Imaging				
	5	NIRCam Wind	NIRCam Imaging	(1) NGC-4945
	6	NIRCam Center	NIRCam Imaging	(1) NGC-4945
NGC4945 NIRSpec				
	7	NIRSpec IFU Mosaic	NIRSpec IFU Spectroscopy	(1) NGC-4945

ABSTRACT

The feedback from starbursts and AGN are the biggest unknowns in galaxy evolution, driving massive galactic winds and curtailing galaxy growth. The physical mechanisms that cause this feedback, mechanical and radiative energy and momentum input by massive star clusters and supermassive

black holes, occur on parsec scales reachable only by JWST on the nearest systems. We propose to use the unparalleled capabilities of JWST to obtain imaging and spectroscopic data on NGC 4945, the nearest Seyfert nucleus, and an example of a hybrid AGN/starburst galaxy hosting a galactic wind. Polycyclic aromatic hydrocarbons (PAHs) are quickly destroyed in hot gas, hence JWST's ability to image PAH bands and measure their abundance provides a unique tracer of the cool phase of the galactic wind, a powerful complement to spectroscopic diagnostics and kinematics. These observations will characterize the structure, composition, and physical state of the gas in the AGN/starburst hybrid and provide crucial constraints on shocks and kinematics. Together with existing data, it will allow us to determine luminosities, ages, and masses of massive star clusters, as well as accretion rate and area of influence of the central AGN. Finally, we will measure the galactic wind cool and photoionized phases, and study the survival of the cool phase using the dust continuum and PAH emission. These data uniquely complement high spatial resolution JWST observations of starburst-only systems by providing critical insight into the role of an AGN in a starburst launching a wind.

OBSERVING DESCRIPTION

We carry out 4 primary types of observations:

1) MRS spectroscopy covering the area of the starburst and young clusters as determined from mm-wave spectroscopy and IR observations, as well as the base of the molecular wind. This requires 2 pointings. We enable simultaneous MIRI imaging in the same filters as described below.WISE images have been used to select background positions which avoid bright emission but are still close to the galaxies. MRS and MIRI imager observations are sequenced, sharing a single MIRI MRS+Imager background.

2) MIRI imaging in 2100W, 1130W, 770W, and 560W, with 1000W as the primary "off PAH" filter for continuum subtraction and 2100W as the "small dust grains" continuum reference. The two sides of the outflow are covered with one foorprint each. The PA constraints are designed to allow a reasonably wide scheduling window. The central region corresponding to the starburst is imaged in a second observation with the SUB128 sub-array in the same filters, to avoid saturation problems on the starburst. The background observations for MIRI are obtained with the simultaneous imaging turned on during the MRS background. For this reason the MRS-MRS background, and MIRI imaging are sequenced.

3) NIRCam imaging in the F187N, F212N, and F335M ([FeII], H2 v=1-0, and 3.3 um PAH) with F140M, F250M, and F444W as "continuum" filters bracketing the observations and enabling continuum color corrections. Full array observations are obtained. The visibility contraints mean the footprint is aligned more-or-less along the major axis of the galaxy, but the coverage along the minor axis is reasonable since the known (ionized and molecular) outflow is small. The PA constraints are designed to allow a reasonably wide scheduling window. The central region corresponding to the starburst is imaged in a second unconstrained observation with the SUB640 array and RAPID readout to avoid saturation problems. Care was

taken to keep the data rate for the large mosaic within the guideline of 0.68 MB/s using a MEDIUM8 readout pattern. No background observation is needed for NIRCam.

4) NIRSpec IFU spectroscopy coverting the area of the starburst and the young clusters. This requires a small mosaic. We use the high resolution gratings to enable measurement of kinematics in wind features. The PA constraints are determined for optimal mapping of the starburst area with a reasonably wide scheduling window. We enable parallel NIRCam imaging for these observations which will mostly cover the stellar halo of NGC 4945

Proposal 10675 - Targets - The Nearest Seyfert Starburst Nucleus and its Galactic Outflow

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]				
(2)	NGC4945-off	RA: 13 07 1.7000 (196.7570833d) Dec: -49 36 3.00 (-49.60083d) Equinox: J2000	Epoch of Position: 2000	
Comments: Category=Calibration Description=[External flat field, Photometric, Spectrophotometric]				

Observation

Proposal 10675, Observation 1: MIRI Imaging Outflows

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Comments: full array imaging along the minor axis/outflow axis to measure PAH bands in the outflow. PA is constrained to optimize mosaic along outflow axis. Observations will use as background the imager data acquired with the MRS background, so they are linked in a non-interruptable sequence.

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]

Template

Subarray
FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	2	10.0	10.0	15.0	0.0	DEFAULT

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4						DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	60	1	1	Dither 1	4	4	666.01	
2	F770W	FASTR1	60	1	1	Dither 1	4	4	666.01	
3	F1000W	FASTR1	60	1	1	Dither 1	4	4	666.01	
4	F1130W	FASTR1	60	1	1	Dither 1	4	4	666.01	
5	F2100W	FASTR1	60	1	1	Dither 1	4	4	666.01	

Special Requirements

Group Visits within 53.0 Days
Aperture PA Range 104.83544897 to 114.83544897 Degrees (V3 100.0 to 110.0)
Visits Same PA

Group Observations 1, 2, 3, 4, Non-interruptible

Observation

Proposal 10675, Observation 2: MIRI Imaging Center

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Comments: fast subarray imaging in the center of the starburst to prevent saturation effects. Observations will use as background the imager data acquired with the MRS background, so they are linked in a non-interruptable sequence

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]

Template

Subarray
SUB128

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	2	20.0	20.0	0.0	0.0	DEFAULT

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				1	1	EXTENDED SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F560W	FASTR1	5	1	1	Dither 1	4	4	2.381	
2	F770W	FASTR1	5	1	1	Dither 1	4	4	2.381	
3	F1000W	FASTR1	5	1	1	Dither 1	4	4	2.381	
4	F1130W	FASTR1	5	1	1	Dither 1	4	4	2.381	
5	F2100W	FASTR1	5	1	1	Dither 1	4	4	2.381	

Special Requirements

Group Observations 1, 2, 3, 4, Non-interruptible

Observation

Proposal 10675, Observation 3: MIRI MRS Center

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Comments: covering submm clusters in starburst center. Observations will use MRS background observation, so they are linked in a non-interruptable sequence.

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
1	2	10.0	10.0	0.0	-15.0	DEFAULT

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
3		IMAGER	F1130W	FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	

Special Requirements	
	Aperture PA Range 100.0 to 110.0 Degrees (V3 100.0 to 110.0) Group Observations 1, 2, 3, 4, Non-interruptible

Observation

Proposal 10675, Observation 4: MRS Background

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Comments: background observation usable for MRS and for the imager, far away from the galaxy but within existing 8um IRAC image, no bright sources/emission

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	NGC4945-off	RA: 13 07 1.7000 (196.7570833d) Dec: -49 36 3.00 (-49.60083d) Equinox: J2000	Epoch of Position: 2000	

Comments:
Category=Calibration
Description=[External flat field, Photometric, Spectrophotometric]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
1	SHORT(A)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
2		IMAGER	F770W	FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
2	MEDIUM(B)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	
3		IMAGER	F1130W	FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSLONG		FASTR1	30	1	1	Dither 1	4	4	333.005	
3	LONG(C)	MRSSHORT		FASTR1	30	1	1	Dither 1	4	4	333.005	

Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible

Observation

Proposal 10675, Observation 5: NIRCam Wind

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(NIRCam Wind (Obs 5)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]

Template

Module	Subarray	Target Placement
ALL	FULL	Module gap (large extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULLBOX	2TIGHTGAPS	STANDARD		4

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F187N	F250M	MEDIUM8	4	1	8	8	3263.978	
2	F212N	F335M	MEDIUM8	4	1	8	8	3263.978	
3	F140M	F444W	MEDIUM8	4	1	8	8	3263.978	

Special Requirements

Aperture PA Range 109.92542306 to 124.92542306 Degrees (V3 110.0 to 125.0)

Observation

Proposal 10675, Observation 6: NIRCam Center

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]

Template

Module	Subarray
B	SUB640

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	3	STANDARD		4

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F187N	F250M	RAPID	6	1	12	12	351.856	
2	F212N	F335M	RAPID	6	1	12	12	351.856	
3	F140M	F444W	RAPID	6	1	12	12	351.856	

Observation

Proposal 10675, Observation 7: NIRSpec IFU Mosaic

Mon May 11 21:00:25 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Coordinated Parallel Template(s): NIRCam Imaging

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-4945	RA: 13 05 27.5000 (196.3645833d) Dec: -49 28 4.80 (-49.46800d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Active galactic nuclei, Seyfert galaxies, Starburst galaxies]

Template

NIRSpec IFU Spectroscopy	NIRCam Imaging
TA Method: NONE	Module: B
HFF Readout Mode: false	Subarray: FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	1	10.0	10.0	-15.0	0.0	DEFAULT

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	LARGE	1	4	

Spectral Elements

NIRSpec IFU Spectroscopy	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	NRS	12	1	false	true	NONE	4	4	2104.407	
2	G140H/F100LP	NRS	12	1	true	true	NONE	4	4	2104.407	
3	G235H/F170LP	NRS	12	1	false	true	NONE	4	4	2104.407	
4	G235H/F170LP	NRS	12	1	true	true	NONE	4	4	2104.407	
5	G395H/F290LP	NRS	12	1	false	true	NONE	4	4	2104.407	
6	G395H/F290LP	NRS	12	1	true	true	NONE	4	4	2104.407	

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F090W	F250M	MEDIUM2	5	1	4	4	1803.777	
2	F150W	F250M	MEDIUM2	5	1	4	4	1803.777	
3	F090W	F430M	MEDIUM2	5	1	4	4	1803.777	
4	F150W	F430M	MEDIUM2	5	1	4	4	1803.777	
5	F090W	F277W	MEDIUM2	5	1	4	4	1803.777	
6	F150W	F277W	MEDIUM2	5	1	4	4	1803.777	

Special Requirements	
	Aperture PA Range 58.97164917 to 68.97164917 Degrees (V3 280.0 to 290.0) Aperture PA Range 238.97164917 to 248.97164917 Degrees (V3 100.0 to 110.0) No Parallel Attachments